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PROJECT DOCUMENTATION

Project Title: HDFS Commands

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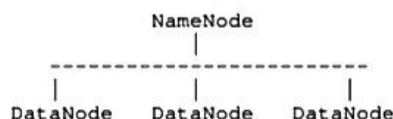
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Introduction: HDFS (Hadoop Distributed File System) is a distributed storage system used to store large amounts of data across multiple machines. It is fault-tolerant and scalable. It stores data in multiple nodes and ensures reliability.

Architecture Diagram:



NameNode manages metadata, while DataNodes store actual data blocks.

Tools / Components Used: Hadoop, Java, Linux, Command Line Interface, Sample Files

Explanation:

1. File System Navigation: It is used to move through directories and manage folders in HDFS. In mobile phone, opening file manager and navigating between folders is similar. Commands like `-ls`, `-mkdir` help to list and create directories. Example: Creating a new folder in phone is similar to creating directory in HDFS.

2. File Operations: It includes uploading, downloading, and viewing files. In mobile phone, uploading photos to cloud or downloading files is similar. Commands like `-put`, `-get`, `-cat` are used. Example: Uploading a file is like sending photo to Google Drive.

3. File Management: It includes copying, moving, and deleting files. In mobile phone, copying or deleting files from storage is similar. Commands like `-cp`, `-mv`, `-rm` are used. Example: Moving a file from one folder to another in phone.

4. Disk Usage: It is used to check storage information like file size and available space. In mobile phone, checking storage usage in settings is similar. Commands like `-du`, `-df` are used. Example: Checking how much storage is used in phone.

5. Permissions: It is used to control access to files. In mobile phone, giving app permissions or locking apps is similar. Commands like `-chmod`, `-chown` are used. Example: Restricting access to files for security.

2. File System Navigation

- `hdfs dfs -ls /path` : Lists files and directories in the given path.

Example: `hdfs dfs -ls /user/data`

- `hdfs dfs -ls -R /path` : Lists all files recursively.

- `hdfs dfs -mkdir /path` : Creates a directory.

Example: `hdfs dfs -mkdir /user/newfolder`

- `hdfs dfs -mkdir -p /path/subdir` : Creates nested directories.

- `hdfs dfs -rmdir /path` : Removes an empty directory.

3. File Operations

- `hdfs dfs -put localfile /hdfs/path` : Uploads file from local system to HDFS.

Example: `hdfs dfs -put data.txt /user/data`

- `hdfs dfs -get /hdfs/file localpath` : Downloads file from HDFS to local system.

- `hdfs dfs -cat /file` : Displays file content.

- `hdfs dfs -head /file` : Shows first few lines of file.

- `hdfs dfs -tail /file` : Shows last few lines of file.

4. File Management

- `hdfs dfs -cp /source /destination` : Copies file within HDFS.

- `hdfs dfs -mv /source /destination` : Moves file.

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- `hdfs dfs -rm /file` : Deletes file.

- `hdfs dfs -rm -skipTrash /file` : Permanently deletes file.

5. Disk Usage & Information

- `hdfs dfs -du /path` : Shows file size.

- `hdfs dfs -df -h` : Displays disk usage in human-readable format.

- `hdfs dfs -count /path` : Shows number of files, directories, and bytes.

6. Permissions

- `hdfs dfs -chmod 777 /path` : Changes permissions.

- `hdfs dfs -chown user:group /path` : Changes owner.

- `hdfs dfs -chgrp group /path` : Changes group.

7. Conclusion

HDFS commands help in managing files and directories efficiently in a distributed environment. Understanding these basic commands is essential for working with Hadoop systems.